JOHNSON ELECTRIC

Testing Manual for Brushless Actuator

Edited by: Zhengchun Zhu, Zheqi Wang, Yulin Liu, Kaiwang Li

Version: V2.1

Date: 2025/12/08

Contents

1. Introduction 1
   1.1 Purpose 1
   1.2 Background and General Information 1
   1.3 Safety and Cautions 1
   1.4 Summary of Test Items 2
   1.5 Sample Appearance and Drawing 2
2. Electrical Characteristics Test 4
   2.1 Speed constant (Kv) 4
      2.1.1 Method 4
      2.1.2 Result 4
   2.2 Line to line resistance 4
      2.2.1 Method 4
      2.2.2 Result 4
   2.3 Line to line inductance 4
      2.3.1 Method 4
      2.3.2 Result 4
3. Performance Test 5
   3.1 Actuator performance test 5
      3.1.1 Method & Process 5
      3.1.2 Result 6
4. Thermal Characteristic Test 9
   4.1 Thermal resistance from shell to ambient 9
      4.1.1 Method 9
      4.1.2 Process 9
      4.1.2 Result 10
   4.2 Temperature Rise Under Load Test 11
      4.2.1 Method 11
      4.2.2 Process 11
      4.2.3 Result 12
   4.3 Temperature Rise Under Overvoltage Test 13
      4.3.1 Method 13
      4.3.2 Process 13
      4.3.3 Result 13

1.Introduction

1.1 Purpose

This document defines the test items and methods for the brushless actuator to ensure that its performance, reliability, and safety conform to design specifications and customer requirements.

1.2 Background and General Information

- Project No.: 25H9012
- Sample Type: Dexterous Hand BLDC Actuator
- Sample Model: HBL-12/1.50-P-70
- Test Environment Requirements:
  a. Ambient Temperature: Normal room temperature (approximately 20 - 25 ℃)
  b. Relative Humidity: Normal indoor humidity (approximately 40 - 60% RH)
- Sample Basic Technical Information：

Rated voltage	12.0 V
Gear ratio	70.2
No load speed	625 rpm ± 15%
Operation tempreture range	10 ℃~ 100 ℃

Table1.1 Basic technical information of BLDC actuator

1.3 Safety and Cautions

For safe use, the actuator should be operated with caution mentioned here. This section provides mandatory safety guidance to be observed before installation, commissioning, and operation. It defines prohibited uses and sets out warnings, as well as general notices and best-practice recommendations.

The listed content applies to users and qualified service personnel; it must be read, understood, and retained for future reference prior to handling the product. Failure to follow these instructions may result in personal injury, equipment damage, or loss of function.

- Operating under high current (Stall current level) longer than described time (50ms) is not recommended.
- Using appropriate rigid coupling for connecting test fixtures to actuator output end is highly recommended.
- FPC bend angle cannot be smaller than 15°.
- Twisting FPC over 180° is prohibited.
- Disconnect and verify zero voltage at the prior or end of each test is recommended.
- Pay attention to the connector orientation during assembly shown below in Figure 1.1.

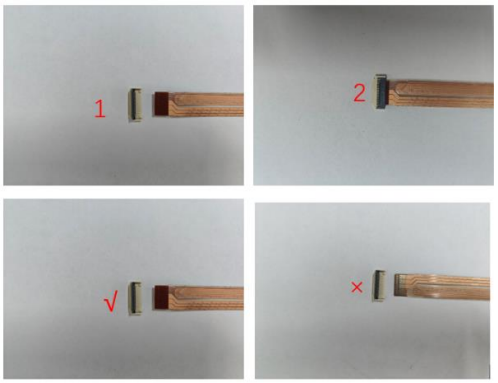


Figure 1.1 Connector orientation reference

1.4 Summary of Test Items

Test Item	Test Content	Remark
Electrical characteristics test	Speed constant (Kv)	Required, detail see 2.1
	Line to line resistance	Required, detail see 2.2
	Line to line inductance	Required, detail see 2.3
Performance test	Performace curves	Required, detail see 3.1
Thermal characteristics test	Thermal resistance to ambient	Required, detail see 4.1
	Temperature rise under load test	Required, detail see 4.2
	Temperature rise under overvoltage test	Required, detail see 4.3
Noise test	Actuator noise under no-load	Optional
Endurance test	Actuator lifespan under load	Optional

Table1.2 Summary of test items

1.5 Sample Appearance and Drawing

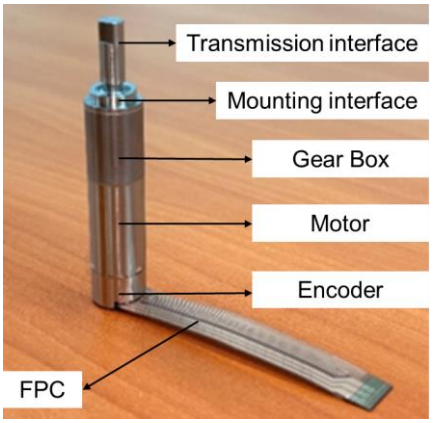


Figure 1.2 Image of BLDC actuator

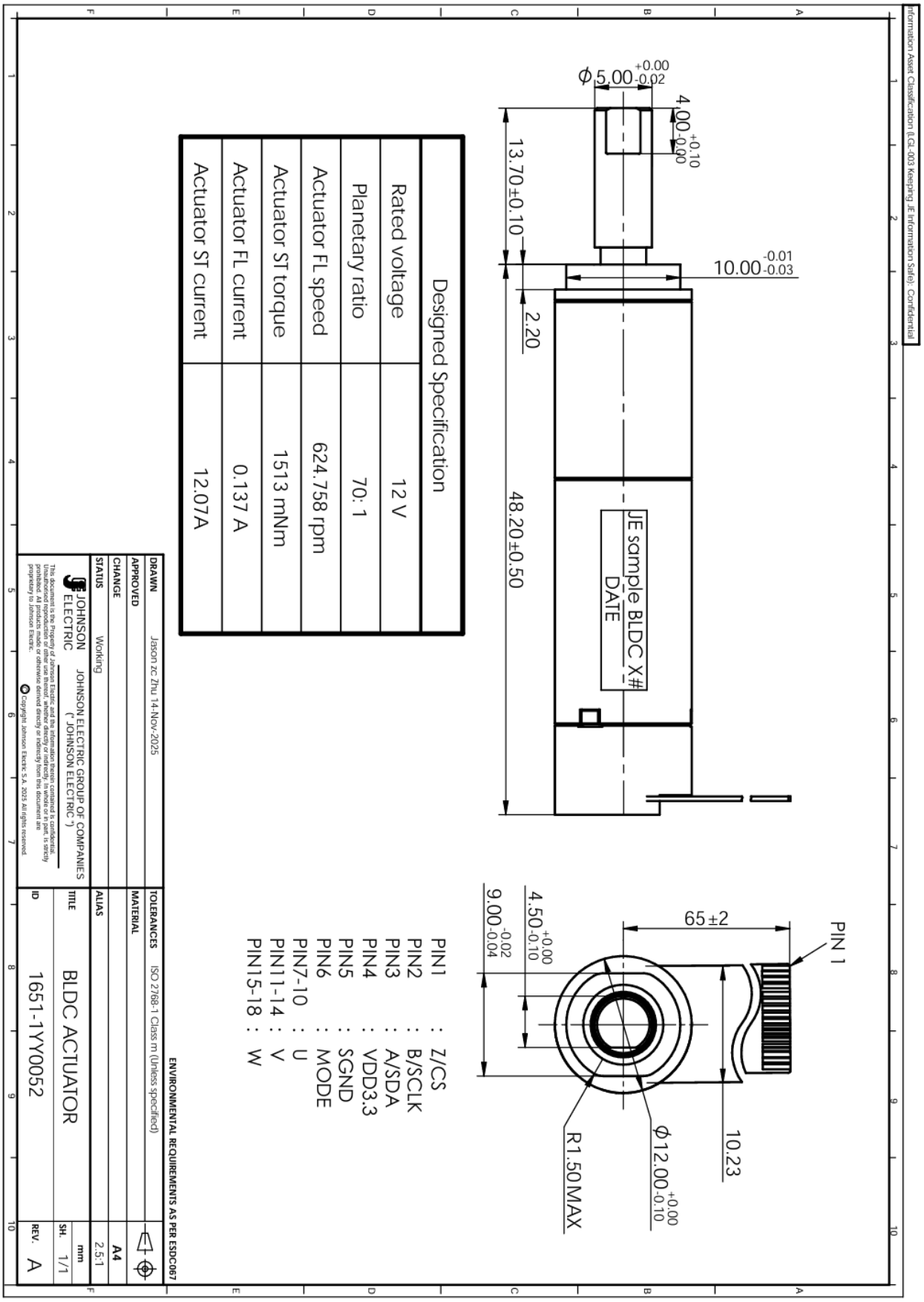


Figure 1.3 BLDC actuator engineering drawing

2. Electrical Characteristics Test

2.1 Speed constant (Kv)

2.1.1 Method

To measure the speed constant of the actuator, It is necessary to record the no-load operating speed of the actuator at the rated voltage.

Calculate the speed constant (Kv) as the ratio of the recorded no-load speed to the rated voltage.

$$k_v = \frac{n\ (free\ load\ speed)}{U\ (rated\ voltage)}$$

2.1.2 Result

Test Item	Unit	Value
No load speed	rpm	650 ± 10%
Speed constant	rpm/V	54.2 ± 10%

Table 2.1 Result of the test of speed constant

2.2 Line to line resistance

2.2.1 Method

The DC resistance between any two-phase terminals (U-V, V-W, W-U) shall be measured at room temperature using a multimeter.

2.2.2 Result

Phase	Unit	Value
U-V	Ω	approx. 0.915
V-W	Ω	approx. 0.915
W-U	Ω	approx. 0.916

Table 2.2 Result of the test of line to line resistance

2.3 Line to line inductance

2.3.1 Method

The inductance between any two-phase terminals (U-V, V-W, W-U) shall be measured at room temperature using a LCR meter at certain frequency.

2.3.2 Result

Phase	Unit	Value
U-V	μH	approx. 23.7
V-W	μH	approx. 20.5
W-U	μH	approx. 20.4

Table 2.3 Result of the test of line to line inductance

3. Performance Test

3.1 Actuator performance test

3.1.1 Method & Process

This test involves obtaining the actuator's characteristic curves for torque-speed, efficiency, current, and power.

The actuators are securely mounted on the test bench. A controller board is used to provide the input voltage and operate the actuator. Simultaneously, a dynamometer applies load to the actuators. Starting from a no-load condition, the load torque is increased gradually and linearly according to predefined test points. Measurements are taken at three specified load points: 50 mNm, 100 mNm, and 150 mNm.

1. The test system records the torque T and rotational speed n at each test point.
2. The test system records the power P_{in} at each test point.

$$P_{in} = U_{in} \cdot I_{in}$$

U_{in}: Voltage at the input terminal of the drive (unit: V)
I_{in}: Current at the driver input terminal (unit: A)
and record the output mechanical power：

$$P_{out} = \frac{2\pi nnT}{60000}$$

T: Torque (unit: mNm)
n: Speed (unit: rpm)
The efficiency of each workstation is ultimately calculated and obtained:

$$\eta = \frac{P_{out}}{P_{in}} \times 100\%$$

As summary, these curves as follows could be obtained:

Performance curves	Description
Torque - Speed	Demonstrates the relationship between output torque and rotational speed across the entire operating range of the motor.
Torque - Efficiency	Demonstrates the operating efficiency of the actuator under various torque and speed combinations.
Torque - Power	Demonstrates the relationship between output mechanical power and output torque.
Torque - Current	Demonstrates the relationship between actuator input current and output torque (or rotational speed).

Table 3.1 Description of actuator performance test curves

3.1.2 Result

Sample number	No Load speed (rpm)		No load current (A)		Stall torque (mN·m)		Stall current (A)	
	CW	CCW	CW	CCW	CW	CCW	CW	CCW
#1	644	655	0.956	0.828	1527.31	1514.69	14.125	12.293
#2	661	662	1.157	1.097	1556.99	1591.14	13.084	13.461
#3	659	665	0.805	0.720	1553.99	1514.06	12.311	12.413

Table 3.2 Table of actuator performance test data

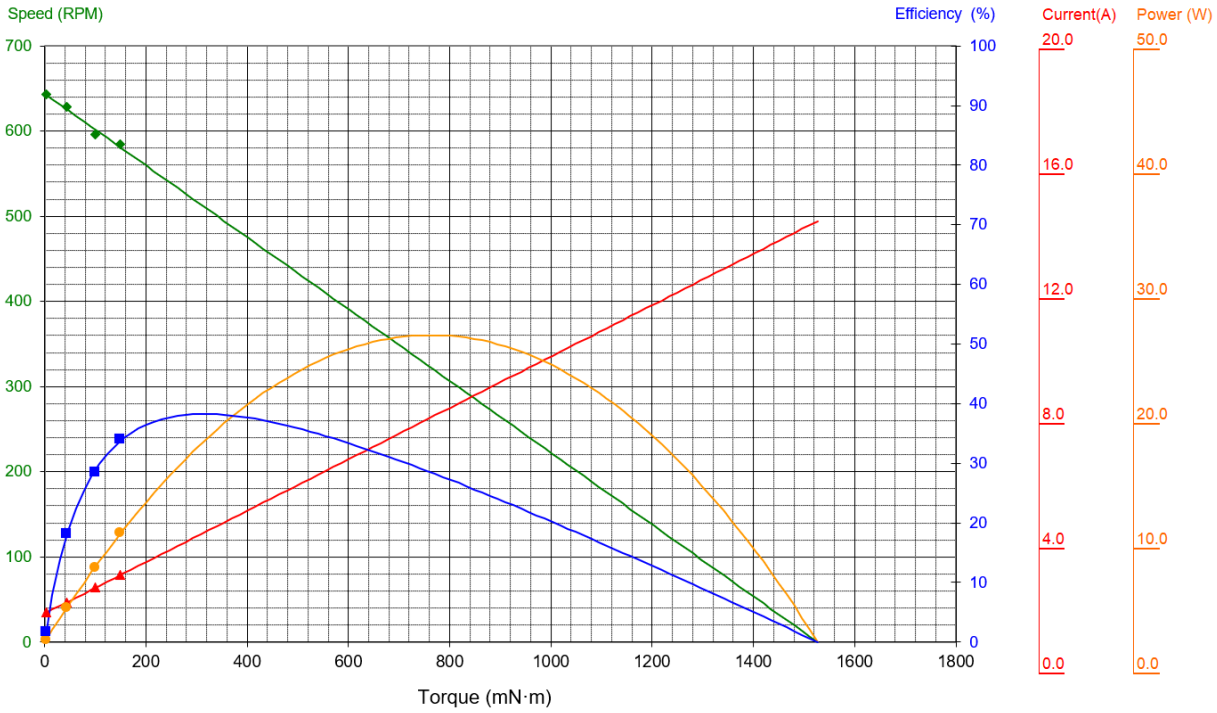


Figure 3.1 Actuator 1# performance curves in CW direction

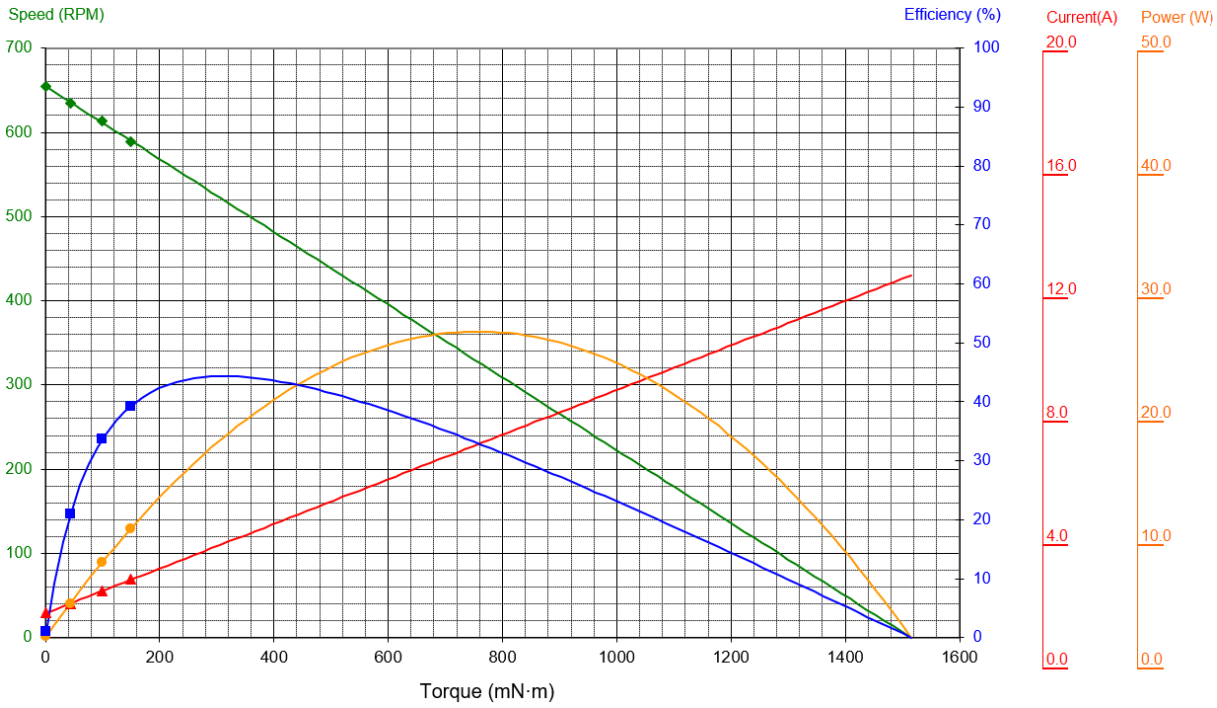


Figure 3.2 Actuator 1# performance curves in CCW direction

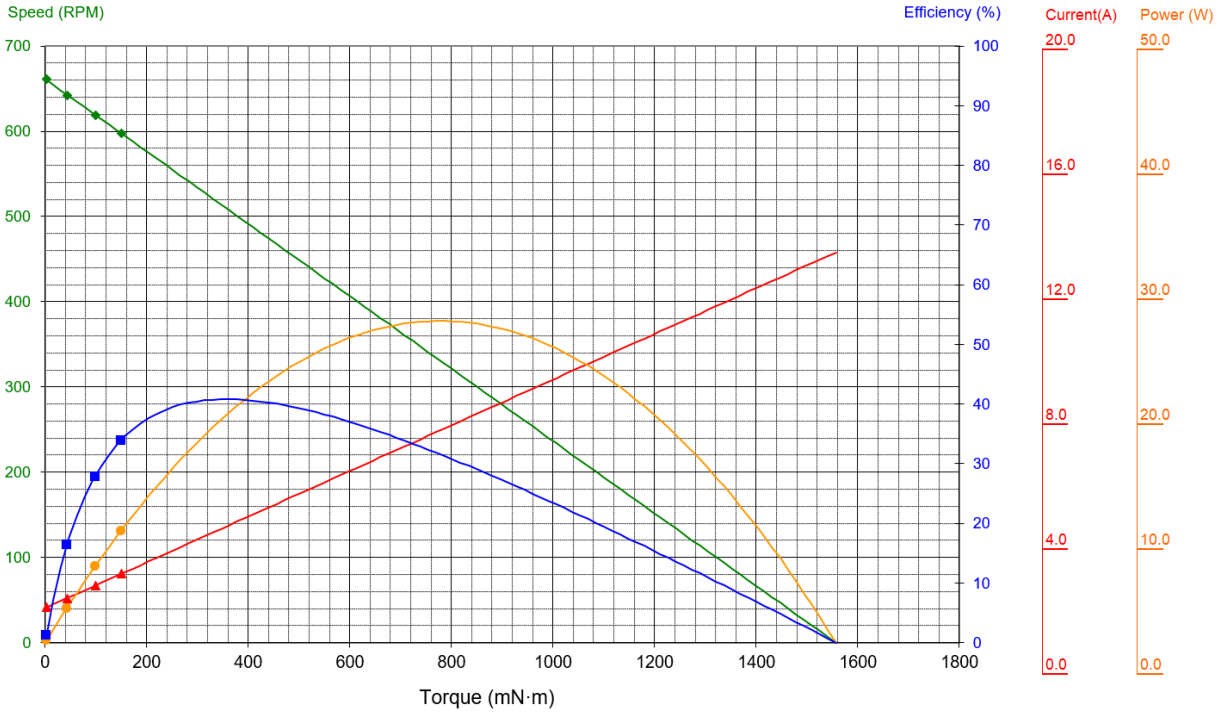


Figure 3.3 Actuator 2# performance curves in CW direction

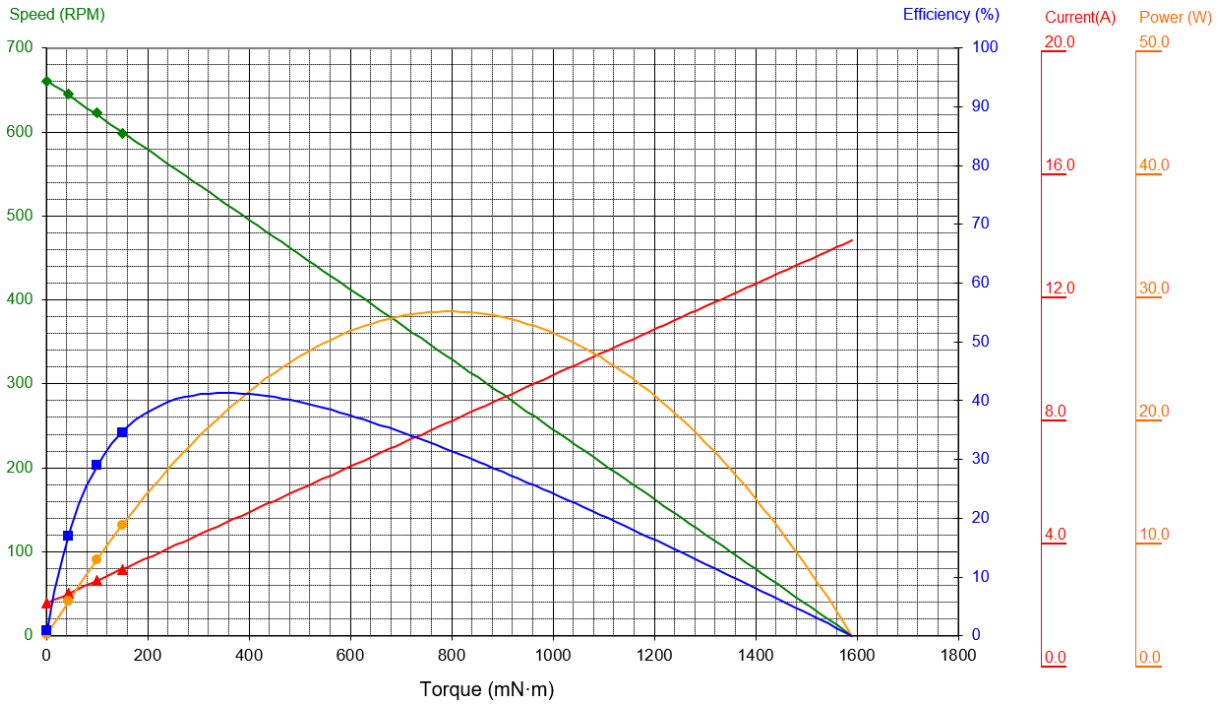


Figure 3.4 Actuator 2# performance curves in CCW direction

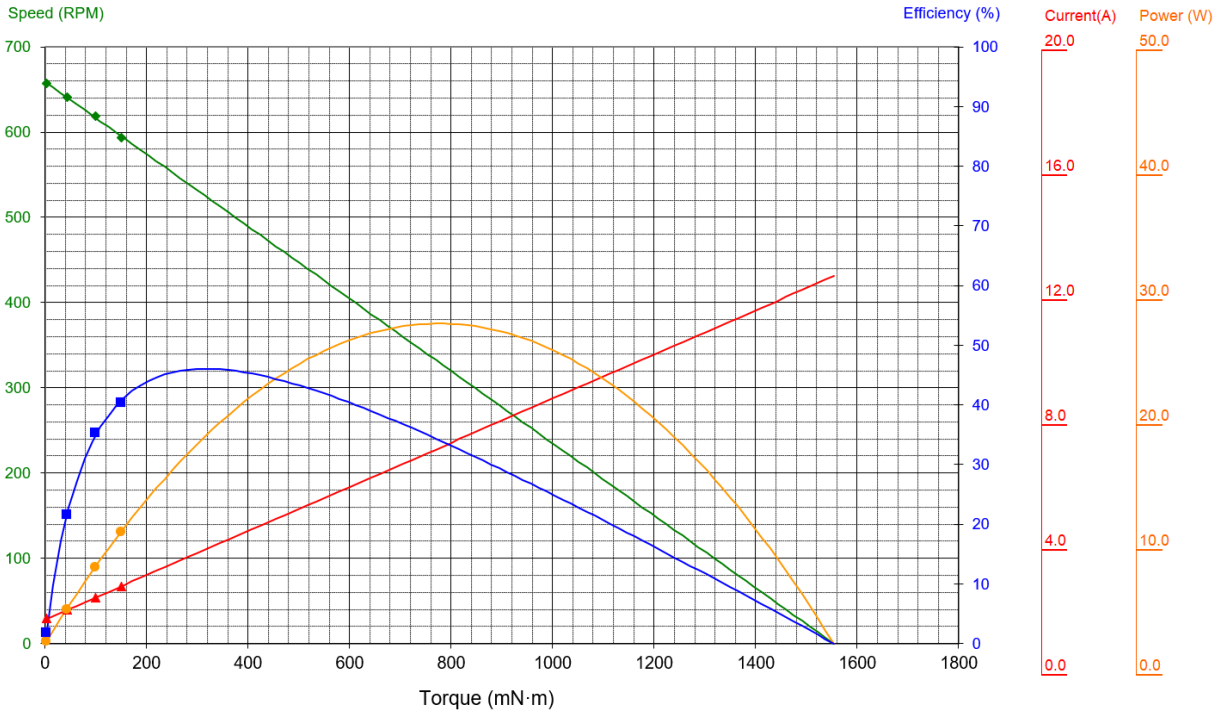


Figure 3.5 Actuator 3# performance curves in CW direction

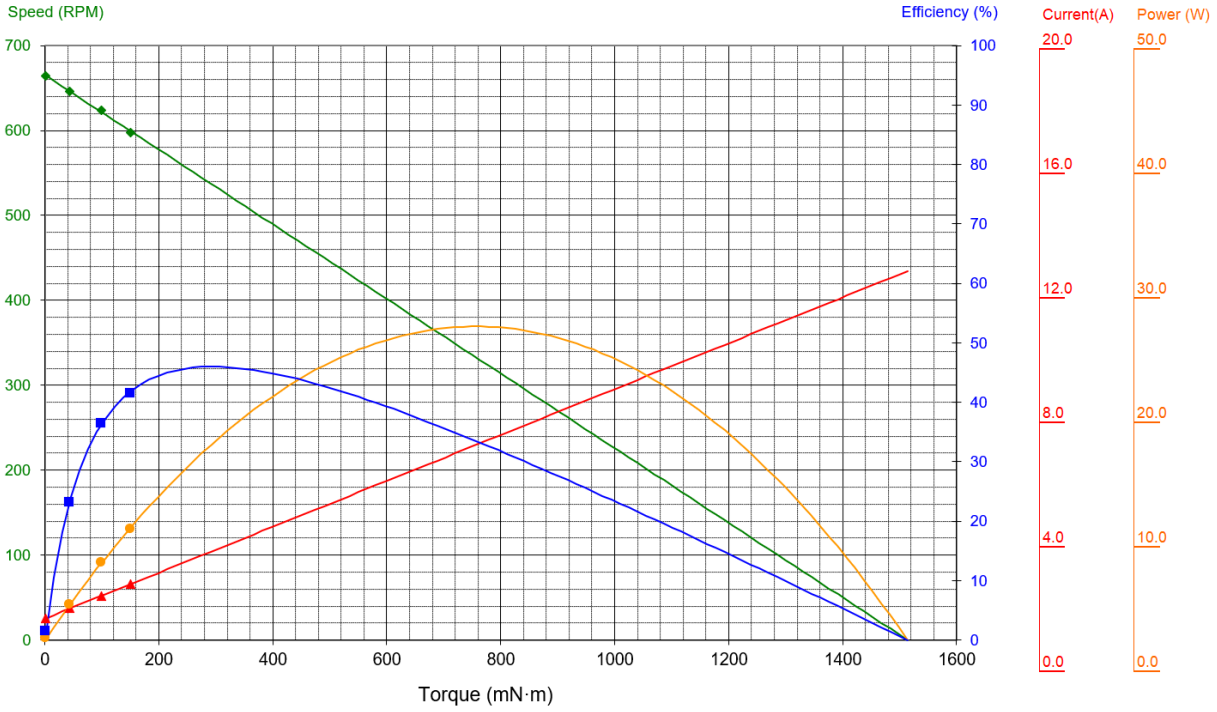


Figure 3.6 Actuator 3# performance curves in CCW direction

4. Thermal Characteristic Test

4.1 Thermal resistance from shell to ambient

4.1.1 Method

According to the equation of calculating motor thermal resistance:

$$R_\theta = \frac{\Delta T}{P_{loss}}$$

R_θ: thermal resistance (unit K/W)

ΔT: temperature difference between the heat source and ambient environment (unit: K)

P_{loss}: total loss (unit: W)

Therefore, the key to the test lies in measuring the heat dissipation power $P_{loss}$and the resulting temperature rise ΔT.

4.1.2 Process

The tested actuator is mounted on a test bench and connected to the load brake via a coupling. Under rated voltage of 12.0V and air cooling (without additional cooling), the load brake is set to a load point of approximately 100mNm. The actuator shall be operated continuously under this load until thermal stability is achieved (determined when the temperature variation at critical housing measurement points does not exceed ±1.0℃ within 5 minutes). During this period, input electrical power is calculated by measuring input voltage and current, while output mechanical power is determined from output speed and torque. Total losses are computed in real time.

Simultaneously, temperatures are monitored via thermocouples (TT-K-30-500) embedded in the main heat dissipation area of the housing, along with their temperature displays, while recording ambient temperature as the baseline. Ultimately, thermal resistance can be calculated. The highest steady-state temperature at the casing measurement point represents the average total loss value under thermal equilibrium conditions.

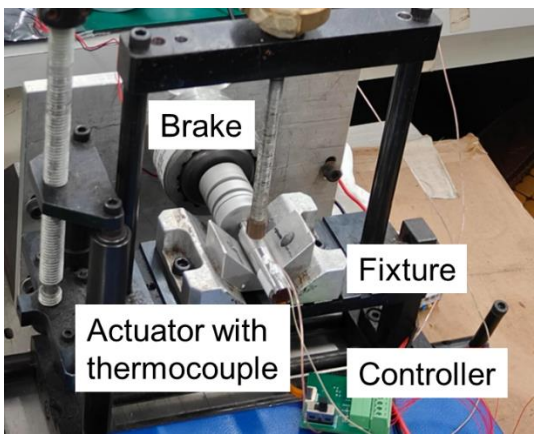


Figure 4.1 Test set-up for measuring thermal resistance of actuator

4.1.2 Result

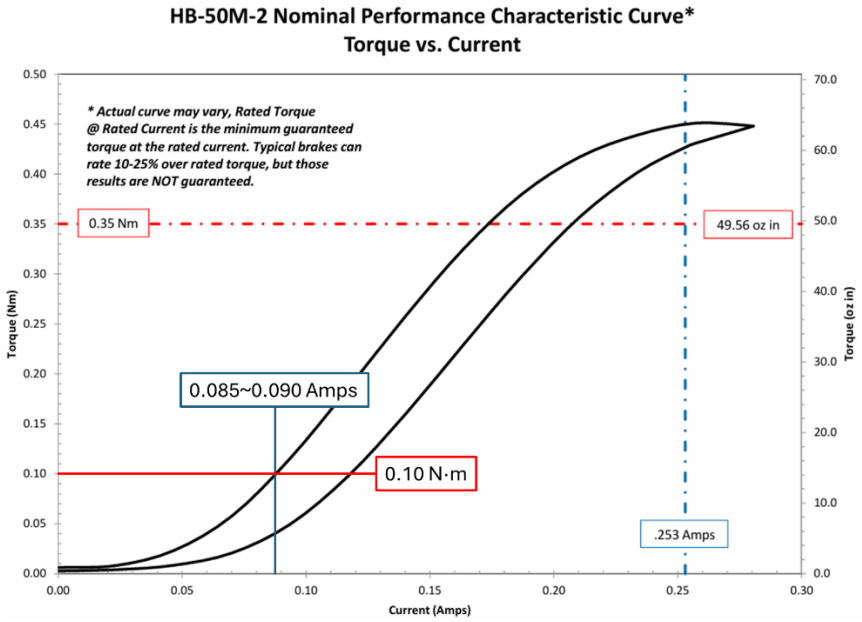


Figure 4.2 Utilized break performance curve [Cited from Magtrol official document]

Under this situation, the parameters shown bleow could be obtained:

Parameters	Unit	Acquired by	Value
Input voltage U_{in}	V	Measured	12.00
Input current I_{in}	Amps		0.80
Output torque T_q	N·m		0.10
Output speed n	Rad/s		66.68
Actuator shell temperature T_c	°C		70.65
Ambient temperature T_a	°C		21.85
Input power P_{in}	W	Calculated	9.60
Output power P_{out}	W		6.67

Table 4.1 Summary of main parameters information of thermal resistance test

Thus, according to the equation the thermal resistance of the actuator under designed working situation could be calculated as:

$$R_\theta = \frac{\Delta T}{P_{loss}} = \frac{T_c - T_a}{P_{in} - P_{out}} \approx 16.66\ K/W$$

4.2 Temperature Rise Under Load Test

4.2.1 Method

This test is requested by the customer: using a torque dynamometer or static torque sensing fixture, stall the motor at the rated voltage for 50ms with 3.0s off time. Measure the torque output vs time using the torque sensor and the temperature rise vs. time for this activity. Stop testing until max temperature conditions or thermal steady state has been reached. Repeat tests using different stall currents, using field-oriented control for current tracking.

4.2.2 Process

The actuator is secured to the test bench and connected to the torque sensor (Magtrol TM 305) via coupling. The other end of the torque sensor is fixed using a fixture, thereby achieving stalled rotation of the actuator. Following a program, the actuator is connected to its rated 12V voltage on this test bench. Stall currents are set at 2.0Amps, 4.0Amps, 6.0Amps, 8.0Amps, 10.0Amps, and 12.0Amps. At each stall current setting, the actuator is controlled to stall for 50ms followed by a 3s pause.

Simultaneously, temperatures are monitored via thermocouples (TT-K-30-500) embedded in the main heat dissipation area of the housing, along with their temperature displays. The test is halted when the actuator temperature reaches the limit of 100°C or when the motor temperature is in stable status (the temperature difference shall not exceed 1°C within five minutes). Finally, the temperature changes and torque variations of the actuator are recorded throughout the entire test duration from start to finish.

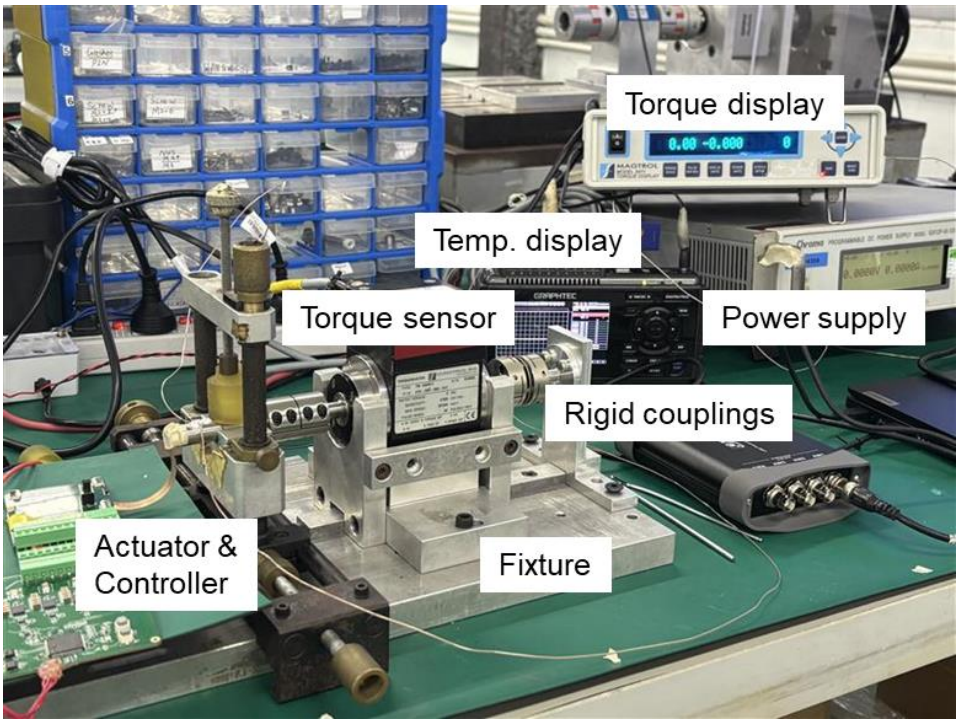


Figure 4.3 Test set-up for load test of actuator

4.2.3 Result

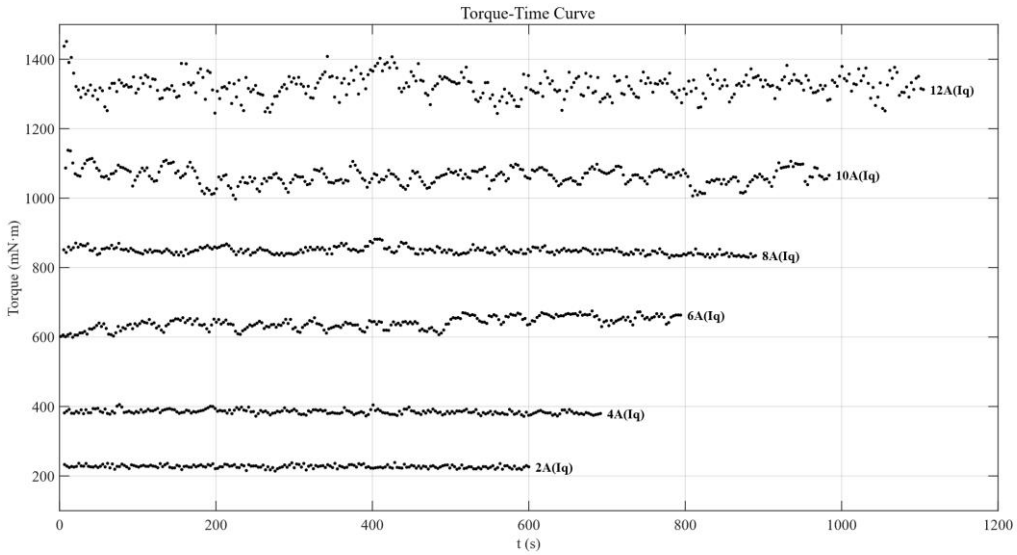


Figure 4.4 Torque – Time curves under different torque currents

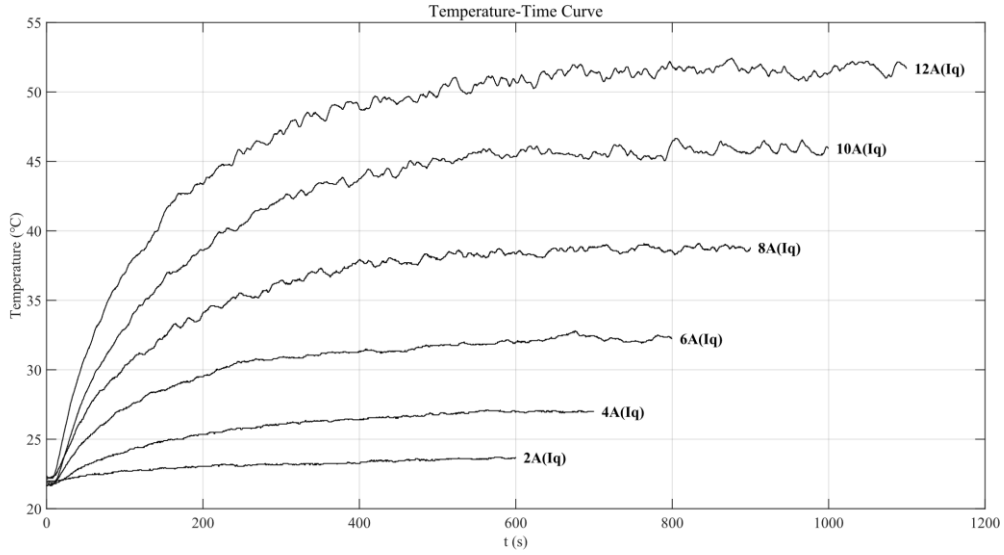


Figure 4.5 Temperature – Time curves under different torque currents

4.3 Temperature Rise Under Overvoltage Test

4.3.1 Method

This test is requested by the customer: run the motor at no-load speed above the rated voltage with a 0.2s on time and 3.0s off time with incrementally increasing drive voltage. Measure temperature rise vs time for this activity.

4.3.2 Process

The actuator is mounted on a test stand without requiring any external mechanical interfaces. By controlling the actuator, it is initiated to run at no load for 0.2s at 12.0V, followed by a 3.0s pause. Upon completion of this cycle, the input voltage needs to be increased by 0.1V, and the actuator shall again run at no load for 0.2s, followed by a 3.0s pause.

Simultaneously, temperature will be monitored via a thermocouple (TT-K-30-500) embedded in the main heat dissipation area of the housing. Record the temperature change of the actuator during the test period as the voltage increases from 12.0V to 18.0V.

4.3.3 Result



Figure 4.4 The mounting position of thermocouple on actuator

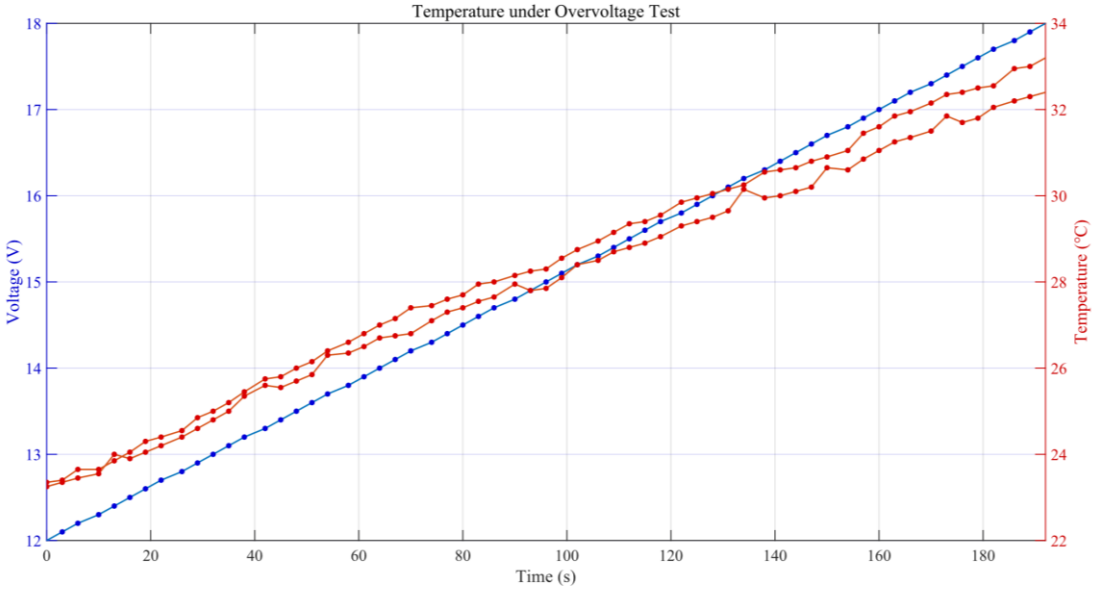


Figure 4.5 Temperature curve under overvoltage test